

Homework 2

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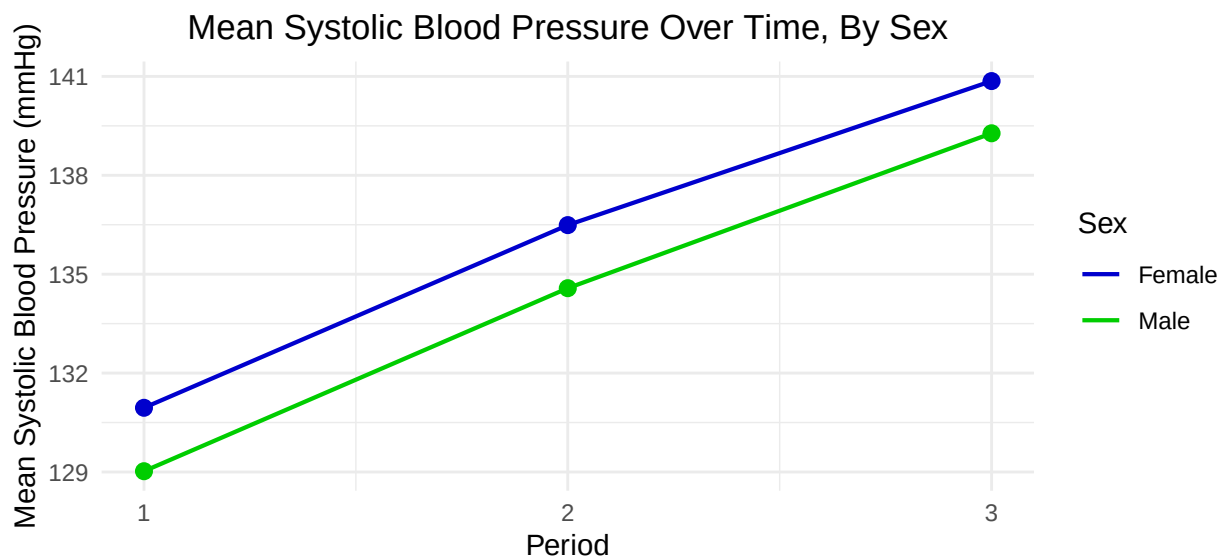
Descriptive Analysis

1. *Descriptive statistics:* Create a table presenting the mean and standard deviation of your continuous outcome at each time point, separately by group.

Systolic Blood Pressure (mmHg) by Period and Sex

Period	Female		Male	
	Mean	SD	Mean	SD
1	130.9	22.1	129.0	16.9
2	136.5	23.2	134.6	19.1
3	140.9	24.2	139.3	21.1

2. *Interaction plot:* Create a plot showing the mean response over time for each group. Based on your plot, comment on whether the profiles appear parallel and whether the groups appear to differ in their overall level.



Based on the above plot, the systolic blood pressure trajectories for males and females appear to be parallel, with both trajectories increasing over time. At each time point, females appear to have higher average systolic blood pressure than males.

Univariate Repeated Measures ANOVA

1. *Sphericity assessment (10 points)*: Assess whether the sphericity assumption is appropriate for your data. Describe your findings and explain how they affect your approach to the within- subject tests.

We used Mauchly's Test of Sphericity which tests if type H covariance structure is valid. After running this test in SAS, we got a p-value of $< .0001$ which tells us that this assumption is not valid for our dataset. To account for this, we will use the adjusted p-values for the within-subject tests.

Sphericity Tests				
Variables	DF	Mauchly's Criterion	Chi-Square	Pr > ChiSq
Transformed Variates	2	0.5907012	1686.2033	<.0001
Orthogonal Components	2	0.9542173	150.10502	<.0001

Figure 1: Mauchly's Test SAS output

2. *Hypothesis tests (20 points)*: Test and interpret the following effects: - Time effect: Is there evidence that the mean response changes over time? - Group effect: Is there evidence that groups differ in their overall mean response? - Group $\times$ Time interaction: Is there evidence that the groups have non-parallel profiles?

For each test, report the relevant test statistic and p-value, and interpret the result in the context of your research question.

For the test of a time effect, we have an F-value of 516.83 and an adjusted p-value of $< .0001$. We have statistically significant evidence that there is relationship between systolic blood pressure over time. This suggests that the lines are not horizontal (have a slope).

For the test of a sex effect, we have an F-value of 7.09 and a p-value of 0.0078. We have statistically significant evidence that there is a relationship between systolic blood pressure and sex. This suggests that males and females have distinct lines (not overlapping).

For the test of an interaction effect, we have an F-value of 0.19 and an adjusted p-value of 0.8179. We do not have evidence that there is an interaction effect between sex and time with relation to systolic blood pressure. This suggests that the lines for sex are parallel.

The GLM Procedure Repeated Measures Analysis of Variance Tests of Hypotheses for Between Subjects Effects					
Source	DF	Type III SS	Mean Square	F Value	Pr > F
SEX	1	7676.911	7676.911	7.09	0.0078
Error	3204	3470173.534	1083.075		

Figure 2: Repeated Measures ANOVA SAS output, sex effect

The GLM Procedure Repeated Measures Analysis of Variance Univariate Tests of Hypotheses for Within Subject Effects							
Source	DF	Type III SS	Mean Square	F Value	Pr > F	Adj Pr > F	
						G - G	H-F-L
sysbp	2	159794.2686	79897.1343	516.83	<.0001	<.0001	<.0001
sysbp*SEX	2	58.5408	29.2704	0.19	0.8275	0.8179	0.8181
Error(sysbp)	6408	990608.9511	154.5894				

Figure 3: Repeated Measures ANOVA SAS output, time and interaction effects

MANOVA

1. *Test of parallelism (10 points)*: Test whether the group profiles are parallel using MANOVA. State what you found and what the result means for your research question.

For the test of parallelism, the Wilks' Lambda value is ≈ 1.00 , and $p = 0.85$. Since the p-value is greater than 0.05, we fail to reject the null hypothesis and conclude that there is no strong evidence for a lack of parallelism. This suggests that there is no interaction effect between sex and time, so the effect of time on mean systolic blood pressure is the same for males and females.

2. *Test of group effect (10 points)*: If appropriate based on your parallelism results, test whether the groups have coincident profiles (i.e., same overall level). Explain why testing for coincident profiles is or is not appropriate given your parallelism results.

Since we concluded above that there is no interaction effect between sex and time, the population mean trajectories for males and for females can be assumed to be parallel. Therefore, it is appropriate to test whether the groups have coincident profiles under the assumption that said profiles are parallel.

For the test of same profiles, $F = 7.09$, and $p = 0.0078$. Since the p-value is less than 0.05, we reject the null hypothesis and conclude that there is evidence of a nonzero effect of sex on mean systolic blood pressure at any fixed time point. This suggests that males and females have parallel but distinct mean systolic blood pressure trajectories.

3. *Test of time effect (10 points)*: Test whether there is an overall time effect using MANOVA. Compare the MANOVA time effect result to the univariate time effect result from Part 2.

For the test of the time effect, $F = 462.31$, and $p < 0.0001$. Since the p-value is less than 0.05, we reject the null hypothesis and conclude that there is evidence of a nonzero effect of time on mean systolic blood pressure, holding sex fixed. This suggests that the mean systolic blood pressure trajectories for both sexes are not flat.

MANOVA Tests for the Hypothesis of No Overall SEX Effect on the Variables Defined by the M Matrix Transformation H = Type III SSCP Matrix for SEX E = Error SSCP Matrix S=1 M=0 N=1600.5		
Statistic	Value	P-Value
Wilks' Lambda	0.99990152	0.8541
Pillai's Trace	0.00009848	0.8541
Hotelling-Lawley Trace	0.00009849	0.8541
Roy's Greatest Root	0.00009849	0.8541

Figure 4: MANOVA SAS output, test of parallelism

MANOVA Test Criteria and Exact F Statistics for the Hypothesis of No Overall sex Effect on the Variables Defined by the M Matrix Transformation H = Contrast SSCP Matrix for sex E = Error SSCP Matrix S=1 M=-0.5 N=1601					
Statistic	Value	F Value	Num DF	Den DF	Pr > F
Wilks' Lambda	0.99779263	7.09	1	3204	0.0078
Pillai's Trace	0.00220737	7.09	1	3204	0.0078
Hotelling-Lawley Trace	0.00221226	7.09	1	3204	0.0078
Roy's Greatest Root	0.00221226	7.09	1	3204	0.0078

Figure 5: MANOVA SAS output, test of sex effect

MANOVA Test Criteria and Exact F Statistics for the Hypothesis of No Overall avg.sex Effect on the Variables Defined by the M Matrix Transformation H = Contrast SSCP Matrix for avg.sex E = Error SSCP Matrix S=1 M=0 N=1600.5					
Statistic	Value	F Value	Num DF	Den DF	Pr > F
Wilks' Lambda	0.77599187	462.31	2	3203	<.0001
Pillai's Trace	0.22400813	462.31	2	3203	<.0001
Hotelling-Lawley Trace	0.28867330	462.31	2	3203	<.0001
Roy's Greatest Root	0.28867330	462.31	2	3203	<.0001

Figure 6: MANOVA SAS output, test of time effect

Comparison and Interpretation

1. Comparison of approaches: Compare the results from the univariate and MANOVA approaches. Did they lead to the same conclusions? If not, explain why the results may differ and which approach you consider more appropriate for your data.

Both approaches yielded the same results. We found that there is consistent evidence of a time and group effect, but no evidence of an interaction effect. Notably the test of a sex effect had the exact same F-stat and p-value. Between these two methods, I believe that MANOVA is more appropriate for our data. While conducting the repeated measures ANOVA, we found that our data does not fulfill the assumption of a Type I covariance matrix. On the contrary, MANOVA does not assume a strict covariance structure so we do not have issues there.

2. Scientific interpretation: Summarize your findings as you would for a collaborator. What did you learn about how your outcome changes over time and whether groups differ?

Our findings suggest that for both males and females, average systolic blood pressure changes over time. More specifically, according to the plot of the population mean trajectories, it appears to increase over time. We have also found that the rate of increase over time is the same for both males and females. However, at each time point, mean systolic blood pressure differs between males and females. According to the plot of the population mean trajectories, mean systolic blood pressure is higher than mean systolic blood pressure for males at each time point.

3. Limitations: Discuss at least two limitations of these classical approaches for your dataset.

MANOVA models the covariance between time periods using an unstructured covariance matrix. While this is a more realistic assumption than the compound symmetry that is assumed in repeated measures ANOVA, interpreting results from the estimated covariance matrix is less straightforward than it is with repeated measures ANOVA. Additionally, both of these methods treat time as a discrete factor and do not explicitly account for the spacing of measurement times. In our study, visits are approximately 6 years apart, however there is variability from patient to patient. The time from baseline to 1st follow up ranges between 1734 and 2562 days (4.75 to 7.02 years) and time from baseline to 2nd follow up ranges between 3899 and 4854 days (10.68 to 13.29 years).

Appendix

```
proc import datafile="/home/u63963979/bios767/data/framingham_wide.csv"
  out=work.framingham
  dbms=csv
  replace;
  getnames=yes; /* Uses first row as variable names */
  guessingrows=max; /* Scans all rows to define column types */
run;

proc glm data = framingham;
  class sex;
  model sysbp1 sysbp2 sysbp3 = sex / noint;
  repeated sysbp / printe nom;
run;
```

Figure 7: SAS Code for Univariate Repeated Measures ANOVA

```

proc import datafile="/home/u63923986/BIOS767/framingham_wide.csv"
  out=work.framingham
  dbms=csv
  replace;
  getnames=yes;
  guessingrows=max;
run;

/* Test for Parallelism */
proc glm data=framingham;
  class sex;
  model sysbp1 sysbp2 sysbp3 = sex / nouni;
  manova h=sex m=(1 -1 0,
                  1 0 -1) / mstat=exact;
run;

/* Test for Same Profiles (Assuming Parallelism) */
proc glm data=framingham;
  class sex;
  model sysbp1 sysbp2 sysbp3 = sex / nouni noint;
  contrast "sex" sex 1 -1;
  manova m=(0.3333 0.3333 0.3333);
run;

/* Test for Time Effect (Assuming Parallelism) */
proc glm data=framingham;
  class sex;
  model sysbp1 sysbp2 sysbp3 = sex / nouni noint;
  contrast "avg.sex" sex 0.5 0.5;
  manova m=(1 -1 0,
            1 0 -1);
run;

```

Figure 8: SAS Code for MANOVA